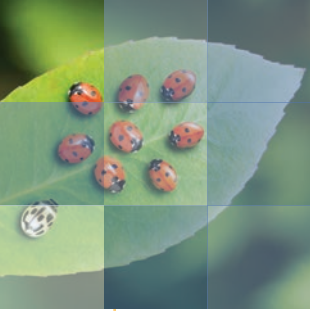




The Normal Distribution

A bell-shaped curve is often used for data analysis and probability.

Identifying Normal Distributions

Definition

A **normal distribution** is bell shaped and is centered around the mean.

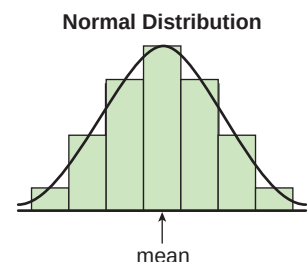
If you were to collect many unopened boxes of the same cereal and weigh them, the weights would all vary slightly but would be centered around the mean weight of all boxes. Some boxes would weigh slightly less than the mean and some slightly more, but most would weigh closer to the mean, creating a bell-shaped curve, or a normal distribution.

Some distributions are not normally distributed. The years on coins, for example, are not normally distributed. Most coins will have more recent years, and few will have older dates. However, no coins can have dates that are greater than the current year, creating a distribution that is shaped more like a triangle. It is not bell shaped.

Using the Normal Distribution

When a data set has a normal distribution, a histogram that represents the data set is symmetric and has a bell shape.

When a curve is drawn through the tops of the bars, the curve is in the shape of a bell.



Normal distributions have important properties that help us study data sets that are otherwise too large to study.

For example, the mean is always at the center of the distribution, so about half of the data set is above the mean and half of the data set is below the mean.

The standard deviation of a normal distribution also gives us useful information about the data set.

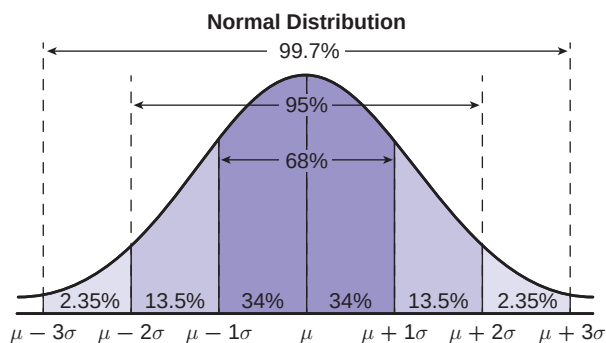
► **Think About It** In a normal distribution, the mean, median, and mode all have approximately the same value.

Properties of a Normal Distribution

If a data set has a normal distribution, then it has the following traits:

- About 68% of the data are within one standard deviation of the mean.
- About 95% of the data are within two standard deviations of the mean.
- About 99.7% of the data are within three standard deviations of the mean.

The properties involving standard deviation are illustrated in more detail in this diagram.



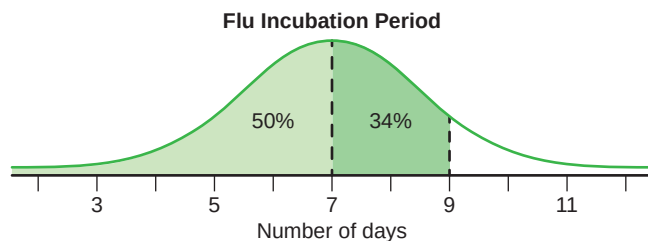
For example, you know that about 68% of the data are within one standard deviation of the mean. This means that about 34% are one standard deviation above the mean and 34% of the data are one standard deviation below the mean.

► **Remember** The Greek letters μ and σ are used to represent the mean and standard deviation of a data set, respectively.

Using Normal Distribution Properties

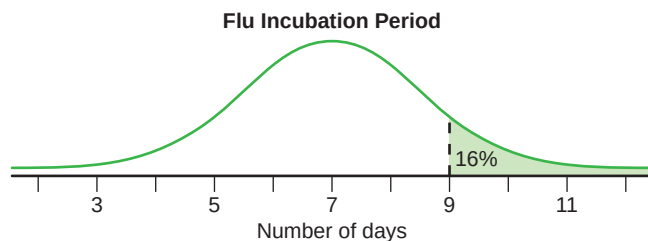
A pediatrician reads an article that states that the length of the flu incubation period is distributed normally with a mean length of 7 days and a standard deviation of 2 days. Afterward, she wonders what the probability is of the flu incubating for at least 9 days.

Assuming that flu incubation periods are normally distributed, a normal curve can be drawn to represent this situation.



Nine days is one standard deviation above the mean of 7 days. So 34% of the data are between 7 days and 9 days. Also, 50% of the data are below the mean.

This means that $50\% + 34\% = 84\%$ of flu cases have an incubation period of at most 9 days. By subtracting 84% from 100%, you can obtain the percent of cases that have an incubation period of at least 9 days.



Therefore, the probability is about 16% that a flu case will have an incubation period of at least 9 days.

► **Remember** In a normal distribution, about 50% of the data are above the mean and about 50% of the data are below the mean.